We remark that the finite-limit theorem also holds in the coordinatewise projective line $(\mathbb{P}^1(\mathrm{VF}))^m$, with the valuation topology: if $A \subseteq \mathrm{VF}^m$ is not bounded, take its closure there and allow infinite coordinates in L. This is the projective version of the same cited theorem; no topological local compactness is assumed.

revision of the next few lemmas

Lemma 2.46 (Curve selection)**.** *Suppose that $\mathbb{S}$ is* VF*-generated. Let $A \subseteq \mathrm{VF}^n$ be definable and let b be a point in its closure in the coordinatewise projective line $(\mathbb{P}^1(\mathrm{VF}))^n$, with the valuation topology. Then there is a b-definable continuous finitary function $g : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ whose values are nonempty and such that*

$$\lim_{\mathcal{M}\setminus\{0\} \longrightarrow 0} g = \{b\}.$$

Resolved referee R165 [Clarification; new] – Definability and nonempty values in curve selection.
Original source line 1081. State that A is definable. In the proof require $g(c) \neq \varnothing$ for every c (or at least on a punctured neighborhood of 0): "finitary" in the earlier definition allows empty values, and $g(c) \subseteq \mathfrak{o}(b, \mathrm{val}(c))$ alone is also satisfied by the identically empty correspondence. The algebraic-closure construction, once its base hypothesis is corrected below, supplies finite nonempty values and hence the stronger conclusion $\lim g = \{b\}$.

Author resolution. The author approves explicitly taking $A \subseteq \mathrm{VF}^n$ definable and requiring finite nonempty values at every input. The construction gives the stronger singleton limit $\{b\}$. Projective charts handle infinite coordinates without changing the affine target set of the curve.

Proof. Use a coordinatewise projective chart around b to reduce to the case $b \in \mathrm{VF}^n$: at an infinite coordinate use inversion and restrict A to the chart. These changes are definable over b and preserve convergence and continuity locally.

Naming the finite tuple b and a nonzero $c \in \mathcal{M}$ preserves VF-generation and the standing nontriviality of the ground valuation. Hence $\mathrm{acl}(\mathbb{S}, b, c)$ is a model of $\mathrm{ACVF}(0,0)$. Since $b \in \overline{A}$, the definable set $A \cap \mathfrak{o}(b, \mathrm{val}(c))$ is nonempty, so elementarity gives a point of that intersection algebraic over $(\mathbb{S}, b, c)$. Compactness supplies a b-definable finitary function g_0 on $\mathcal{M} \setminus \{0\}$ such that

$$\varnothing \neq g_0(c) \subseteq A \cap \mathfrak{o}(b, \mathrm{val}(c))$$

for every c. Thus g_0 converges to $\{b\}$, and nonempty values make this limit minimal.

By Lemma 2.44, with its one-dimensional source, g_0 is continuous outside a finite set. Choose a b-definable clopen disc $D \subseteq \mathcal{M}$ around 0 such that g_0 is continuous on $D \setminus \{0\}$. Also $\mathrm{acl}(\mathbb{S}, b)$ is a model, so A contains a finite nonempty b-definable set E. Define

$$g(c) = \begin{cases} g_0(c), & c \in D \setminus \{0\}, \\ E, & c \in \mathcal{M} \setminus D. \end{cases}$$

The two pieces are relatively clopen in $\mathcal{M} \setminus \{0\}$, so g is continuous, finite-valued and nonempty everywhere. Its limit at 0 is still $\{b\}$. Finally transport the construction back through the projective chart. This avoids assuming that a definable value-group radius is the valuation of a definable scalar. □

Resolved referee R040 [Error] – Curve selection requires a restriction on the base.
Original source line 1088. The assertion is false for the arbitrary $\mathbb{S}$ allowed so far. Let K be a nontrivially valued algebraically closed base field, add an RV parameter t with $\mathrm{vrv}(t) = 0$ transcendental over its residue field, and put $A = \{(x, y) : y \neq 0, \mathrm{rv}(x/y) = t\}$. Then $(0,0) \in \overline{A}$. A nonempty finite $\mathbb{S}$-definable

choice in A at an input $c \in K \cap (\mathcal{M} \setminus \{0\})$ would have coordinates algebraic over K, t, c, hence in K by Lemma 2.13; their ratio cannot have leading term t. Even though finitary values may be empty, $b \in \lim g$ forces their support to accumulate at 0; by C-minimality that support contains a punctured neighborhood, whose defining radius lies in $\Gamma(K)$ since adjoining the residue parameter t does not enlarge definable values. It therefore contains such an input $c \in K$. Thus the claimed curve does not exist over that base. Impose VF-generation before this lemma (as already suggested in the human report), or allow explicitly specified extra VF representatives. Under VF-generation, the added nonzero $c \in \mathcal{M}$ makes the algebraic closure a model, and the displayed construction yields nonempty finite values converging to b itself. For the last rescaling step, first shrink to a disc whose radius is the valuation of a definable scalar; the original definable radius need not be realized by such a scalar.

Author resolution. The author approves VF-generation as a local hypothesis of this lemma, validating the algebraic-closure model argument. The continuous germ is extended by a constant finite nonempty set on a clopen complement, so no definable scalar realizing an arbitrary definable radius is assumed. Later geometric uses may temporarily enlarge the base; no global VF-generation assumption is imposed here.

This is an analogue of the highly useful o-minimal curve selection.

Remark 2.47. Although the Γ-sort is essentially an o-minimal group, curve selection fails due to the presence of the top element ∞. For example, the set $\Gamma \times (0, \infty) \subseteq \Gamma_\infty^2$ contains curves approaching the boundary point $(0, 0)$ or (∞, ∞) but no curves approaching the boundary point $(\infty, 0)$, because every definable one-parameter curve is piecewise rational-affine. Near a one-sided endpoint, one affine piece remains, and a positive affine coordinate cannot tend to 0 while another coordinate tends to ∞.

Resolved referee R166 [Typo; new] – Eventually affine curves in the value group.
Original source line 1094. Replace "all curves are lines" by "every definable one-parameter curve is piecewise rational-affine". Near a one-sided endpoint, one affine piece remains; a positive affine coordinate cannot tend to 0 while another coordinate tends to ∞. This is the precise reason the stated boundary point cannot be approached.

Author resolution. The author approves the precise piecewise rational-affine formulation. An eventual affine piece gives the stated obstruction to approaching $(\infty, 0)$; the example and the conclusion are unchanged.

Lemma 2.48. *Let $A \subseteq \mathrm{VF}^n$ be definable, closed and bounded, and let $f : A \longrightarrow \mathrm{VF}^m$ be definable and continuous. Then $f(A)$ is also closed and bounded.*

Proof. The conclusion is geometric, so we may temporarily enlarge the parameter base to a small VF-generated base over which all the data are definable. The standing nontriviality of the ground valuation is preserved. Over this base, and after naming any additional finite VF tuple, algebraic closure is a model of $\mathrm{ACVF}(0, 0)$. Elementarity and compactness therefore give finite nonempty definable choices in the nonempty families below. No choice over the original parameter base is asserted.

First let $b \in \overline{f(A)} \setminus f(A)$, where this closure is affine, and name b. For every $c \in \mathcal{M} \setminus \{0\}$ the set

$$\{x \in A : f(x) \in \mathfrak{o}(b, \mathrm{val}(c))\}$$

is nonempty. Applying the model argument over the enlarged base together with (b, c), and then compactness, gives a definable finitary function $F : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ such that every $F(c)$ is nonempty and $f(F(c)) \subseteq \mathfrak{o}(b, \mathrm{val}(c))$. Lemma 2.45, in its finite-dimensional form, gives a finite minimal limit set $L \subseteq A$. This set is nonempty, since F has nonempty values arbitrarily close to 0. For every $a \in L$, minimality implies that values of F meet every neighborhood of a for inputs